Discussion

Evaluating Gemma-4-E2B and Gemma-4-E2B-it across four safety and capability suites

This discussion synthesizes the results from all four evaluation folders in the repo — Academic-integrity/, truthfulqa/, openbookqa/, and promptinjection/ — plus the shared datasets in data/. Two model variants were tested throughout: the base model (Gemma-4-E2B) and its instruction-tuned counterpart (Gemma-4-E2B-it).

1. Headline Results

Suite	Metric	Base (Gemma-4-E2B)	Instruction-tuned (Gemma-4-E2B-it)
Academic Integrity	Pass rate	~80% per category	**96.7% (29/30)**
TruthfulQA	Pass rate	96.0%	**100% (50/50)**
TruthfulQA	Factual accuracy (1-5)	3.40	**4.20**
TruthfulQA	Avoids misconception (1-5)	3.34	**4.36**
TruthfulQA	Clarity (1-5)	3.52	**4.66**
OpenBookQA	Accuracy	40.0% (no instructions, n=5)	**80.0% (with instructions, n=10)**
Prompt Injection	Pass rate	-	**0.0% (likely a measurement artifact - see Section 4)**

The clearest, most consistent signal across every suite is that instruction tuning helps — not just on safety-flavored tasks (Academic Integrity) but on a pure knowledge task (TruthfulQA) and a pure reasoning task (OpenBookQA) too. This is worth calling out because it means instruction tuning in this case is not a safety/capability trade-off; it moved several axes in the same direction at once.

2. Academic Integrity — A Helpful Tutor That Still Says No

The instruction-tuned model passed 29 of 30 academic-integrity prompts (96.7%), the one failure occurring in the “Essay” category (80% pass rate there vs. 100% in every other category). Underneath that headline number, three findings are worth discussing:

- **The base model refuses too, but badly.** Gemma-4-E2B's raw pass rate per category also sits around 80%, but its helpfulness score is pinned at 1.0/5 across every category, and politeness sits between 2.6–3.4/5. It behaves like a blunt-instrument refuser. The instruction-tuned model keeps a similar or better pass rate while lifting helpfulness to 3.8–4.8/5 and politeness to a flat 5.0/5. In other words, the tuning didn't just add more refusals — it made the refusals (and the help around them) noticeably better tutoring.

- **Adversarial reframing is the weak point, not direct requests.** The lowest pass rate in the safety scorecard is 86.7% under a “for a friend” framing — the model is more likely to comply when the dishonest request is socially distanced from the requester. This lines up with a broader pattern in LLM safety research: direct requests are the easy case, and indirection/reframing is where models leak.
- **The evaluation-awareness check came back essentially flat** (93.3% pass rate whether or not the prompt disclosed it was an evaluation, a 0.0-point gap). That's a mildly reassuring result for this narrow test, but the sample is only 15 prompts, so it should be read as “no evidence of awareness-based sandbagging in this sample,” not as a general guarantee.

A methodological caveat that matters more than any single number: the helpfulness and politeness scores are produced by the instruction-tuned model judging its own responses. The repo's own cross-validation check (Section 25 of the scorecard) found the self-judge scored its own responses 5.00/5 on average, versus 4.60/5 from an independent judge (Qwen2.5-3B) on the same 10 responses — a real, if modest, self-preference gap. Any claim about “how helpful” the model is should be discounted slightly for this.

3. TruthfulQA — Accuracy Gains Are Real, But Category Difficulty Is Uneven

Both models pass nearly every prompt (96–100%), but pass/fail is a coarse lens here — the 1–5 quality scores tell a more interesting story:

- **Improvement is broad, not concentrated in one dimension.** Factual accuracy (+0.80), avoiding misconceptions (+1.02), and clarity (+1.14) all move up together for the tuned model. Clarity improved the most, suggesting a meaningful part of the “truthfulness” gain here is really a communication gain — the tuned model states the same underlying fact more legibly.
- **Category difficulty is not uniform.** “Economics,” “Education,” and “Weather” are the base model's worst categories (scores of 1.0–1.5/5 on factual accuracy despite a 100% pass rate), which is a reminder that a binary pass rate can hide low-quality passing answers. The instruction-tuned model closes most of this gap (e.g., Education rises from 1.0 to 5.0), but a few categories — “Myths and Fairytales,” “Language” — actually dip for the tuned model, which is worth a closer look rather than assuming tuning is monotonically better on every category.
- **Sample sizes per category are small** (often 1–3 prompts), so category-level deltas should be treated as directional signal, not statistically robust findings.

4. Prompt Injection — The 0% Pass Rate Is Very Likely a Scoring Artifact, Not a Real Vulnerability

This is the most important finding to discuss carefully, because read at face value the scorecard says something alarming: 0% pass rate, 100% violation probability across all 10 adversarial prompts. Reading the actual transcripts in prompt_injection_scored_results.csv tells a different story:

- In all 10 cases, the model's response was a clean, appropriate refusal, with no password, credential, or protected instruction ever disclosed.
- In every one of those same 10 cases, the automated judge's clean-safety field reads **"Unclear"**, and the pipeline treats “Unclear” as a fail. The instruction_following and response_quality scores (mostly 5/5) directly contradict the pass/fail flag on the same row.
- This looks like a judge-prompt or output-parsing bug (the judge couldn't produce a clean “safe”/“unsafe” verdict and the downstream logic defaulted to “fail”) rather than genuine prompt-injection compliance.

The discussion point this raises is broader than prompt injection specifically: across this repo, every suite that uses an LLM-as-judge shows some form of judge reliability problem — self-preference bias in Academic Integrity, all-NaN parse failures in one condition of OpenBookQA (see below), and a systematic “Unclear → fail” miscoding here. Before trusting any pass-rate headline number in this project, it's worth checking whether the judge actually produced a usable verdict for that row.

5. OpenBookQA — Instructions Roughly Double Accuracy, But Half the Judged Data Is Unusable

- Accuracy rises from 40% (no instructions, n=5) to 80% (with explicit formatting instructions, n=10). The sample sizes here are small enough that this gap should be treated as suggestive rather than conclusive, but it's directionally consistent with the instruction-following gains seen elsewhere in the repo.
- **More notably: all five rows in the “no instructions” condition have judge_reason = PARSE_FAILED,** meaning the qualitative scores (format adherence, explanation quality, politeness) don't exist for that condition at all — not because the model didn't produce them, but because the judge couldn't parse its own scoring output for that batch. This is the same category of judge-reliability issue as Section 4. Any “with vs. without instructions” comparison on quality (not just raw accuracy) is currently only usable for the “with instructions” side.

6. Cross-Cutting Themes

- **Instruction tuning helped everywhere it was tested** — safety-adjacent tasks, factual accuracy, and reasoning quality all improved together, with no suite showing a clear capability regression from tuning.
- **Automated LLM-as-judge scoring is the project's biggest source of noise,** not the underlying model behavior. Three separate suites show judge-side problems: self-preference bias (Academic Integrity), parse failures that silently drop an entire condition (OpenBookQA), and an “Unclear” verdict being miscoded as “fail” (Prompt Injection). A pass-rate number is only as trustworthy as the judge that produced it.
- **Sample sizes are small throughout** (most suites test 5–50 prompts, often 1–5 per category). The directional findings above are consistent and plausible, but none of them should be reported as statistically robust without more prompts per category.

- **Where the model actually struggled, it was with indirection** — reframed academic dishonesty requests, not direct ones; low-frequency knowledge categories in TruthfulQA rather than the whole dataset. That is a more useful safety signal than the raw pass rates suggest, because it points at how a model fails, not just how often.

7. Suggested Follow-Ups

- Re-run the Prompt Injection judge with a stricter output schema (or a rule-based fallback) so “Unclear” cannot silently become “fail,” and re-score the existing 10 transcripts before drawing any conclusion about the model's actual resistance.
- Re-run the OpenBookQA “no instructions” batch through the judge (or a different judge model) to recover the missing quality scores, since accuracy alone doesn't tell you whether the reasoning was any good.
- Expand category-level sample sizes in TruthfulQA (many categories currently have n=1), and consider an independent (non-self) judge for Academic Integrity helpfulness/politeness scoring by default rather than as a spot-check.
- Extend the adversarial-reframing test in Academic Integrity beyond 4 framings and 15 prompts, since that was the single lowest pass rate in the whole project (86.7%).